

OpenCart (Frontend) – Test Plan

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1. Objective

The objective of this test plan is to validate the functionality, usability, reliability, and performance of the OpenCart Frontend application. The testing effort aims to ensure that critical e-commerce workflows operate correctly and meet user expectations before release.

Key objectives include:

- Verify all customer-facing features function as intended.
 - Identify and report defects early in the testing cycle.
 - Validate integration between frontend modules.
 - Ensure compatibility across supported browsers.
 - Verify security-sensitive workflows such as login, registration, and checkout.
 - Improve overall user experience and application stability.
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2. Scope

This test plan covers testing activities for the OpenCart Frontend website.

Testing will include:

- Functional Testing
- Integration Testing
- Regression Testing
- Cross-Browser Testing
- Performance Testing
- Security Testing (limited to frontend payment and authentication workflows)
- User Acceptance Testing (UAT)

Testing will be conducted in supported browser environments and will focus on validating business requirements and user workflows.

3. Inclusions

The following modules and functionalities are included in testing:

User Registration

- New account creation
- Mandatory field validation
- Password validation
- Navigation controls

Login and Logout

- User authentication
- Forgot Password functionality
- Session management
- Logout functionality

Search

- Product search
- Search result validation
- Keyword matching

Product Compare

- Compare products
- Add/remove products
- Product specifications comparison

Home Page (Your Store)

- Navigation menu
- Category browsing
- Product listing
- Pagination
- Currency selection
- Wish List functionality
- Product comparison
- Shopping Cart access

Product Details Page

- Product images
- Product descriptions
- Reviews and ratings
- Add to Cart
- Add to Wish List
- Compare Products

Other Features

- Shopping Cart
 - Checkout
 - My Account
 - Order History
 - Downloads
 - Contact Us
 - Footer Links
 - Category Navigation
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4. Exclusions

The following items are outside the scope of testing:

- Automation Testing
 - API Testing
 - Database Testing
 - Backend Administration Modules
 - Features introduced after test plan approval
 - Third-party integrations not directly related to frontend functionality
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5. Test Environments

Operating Systems

- Windows 10

Browsers

- Google Chrome (Latest)
- Mozilla Firefox (Latest)
- Microsoft Edge (Latest)
- Safari (Latest)

Environment Requirements

- Apache Web Server
- MySQL Database
- Stable LAN/Internet Connection
- Minimum 4GB RAM
- 3.4 GHz Processor

Configuration Management

- GitHub Repository
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6. Defect Reporting Procedure

Defect Identification

A defect will be reported when:

- Actual results differ from expected results.
- Functional requirements are not met.
- UI/UX issues impact usability.
- Performance issues affect user experience.

Defect Lifecycle

1. Tester identifies defect.
2. Defect is logged in Jira.
3. Severity and priority are assigned.
4. Development team investigates.
5. Fix is implemented.
6. QA performs retesting.
7. Defect status is updated.
8. Regression testing is performed when required.

Defect Severity Levels

- Critical
- Major
- Minor

Defect Tracking Tool

- Jira
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7. Test Strategy

Test Design Techniques

- Equivalence Partitioning
- Boundary Value Analysis
- Error Guessing
- Exploratory Testing
- Requirement-Based Testing

Testing Types

- Functional Testing
- Integration Testing
- Regression Testing
- Cross-Browser Testing
- Performance Testing
- Security Testing
- UAT

Test Execution Process

1. Review requirements.
2. Create test scenarios.
3. Create test cases.
4. Review and approve test cases.
5. Execute test cases.
6. Log defects.
7. Retest fixed defects.
8. Execute regression tests.

9. Prepare test reports.

8. Test Schedule

Activity	Owner	Estimated Duration
Requirement Review	QA Team	2 Days
Test Planning	QA Lead	1 Day
Test Scenario Creation	QA Team	2 Days
Test Case Creation	QA Team	5 Days
Test Execution	QA Team	5 Days
Defect Reporting & Retesting	QA Team	Ongoing
Regression Testing	QA Team	2 Days
UAT Support	QA Team	2 Days
Test Closure	QA Lead	1 Day

9. Test Deliverables

Before Testing

- Test Plan
- Requirement Documents
- Test Scenarios
- Test Cases
- RTM

During Testing

- Test Data
- Defect Reports
- Execution Reports
- Traceability Matrix

After Testing

- Test Summary Report
 - Defect Summary Report
 - Release Notes
 - Test Closure Report
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10. Entry & Exit Criteria

Entry Criteria

- Requirements approved.
- Test Plan approved.
- Test Cases approved.
- Test Environment available.
- Smoke Test completed.
- QA resources assigned.

Exit Criteria

- All planned test cases executed.
 - Critical defects closed.
 - No unresolved high-priority defects.
 - Test Summary Report completed.
 - UAT completed.
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11. Test Execution

Entry Criteria

- Approved Test Cases available.
- Stable build deployed.
- Test environment ready.

Exit Criteria

- Test execution completed.
 - Defects logged and tracked.
 - Execution reports prepared.
 - Regression testing completed.
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12. Test Closure

Entry Criteria

- Testing completed.
- Defect verification completed.
- Reports available.

Exit Criteria

- Test Summary Report approved.
- Closure report completed.
- Project sign-off received.

13. Tools

Purpose	Tool
Test Case Management	Microsoft Excel
Test Execution Tracking	Microsoft Excel
Defect Tracking	Jira
Reporting	Excel, Jira
Configuration Management	GitHub

14. Risks and Mitigations

Risk	Mitigation
Payment failures due to network issues	Test under multiple network conditions
Browser compatibility issues	Perform cross-browser testing
Environment instability	Maintain backup environments
Resource unavailability	Cross-train QA team members
Requirement changes	Update RTM and test cases promptly

15. Approvals

Role	Approver	Status
Project Manager	TBD	Pending
QA Lead	TBD	Pending
Stakeholder	TBD	Pending

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